Cyclohexane Reactor Ver 1 · operator training simulator

Document register

Document no. CHX-V1-PM-DOC-001 · **Discipline** Project (document control)

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• 18 documents of the EPC teaching set (334 pages), "Issued for teaching" at Rev 0 (PM-DOC-001 Rev 1, PE-RLB-001 Rev 1), with discipline, page count, source tab and what it is built from.
• 81 engineering queries (29 safety, 25 design, 27 documentation) raised by 9 tabs, each with owner and proposed resolution: 75 open, 1 partly closed, 5 closed (EQ-er-10, EQ-la-2, EQ-la-3, EQ-la-8, EQ-ej-1, EQ-ej-2).
• 83 open recommendations (HAZOP 38, Safety layers 44, LOPA 1) with the documents each must change.
• 31 checks across the document set: 17 pass, 14 fail; every failure is an engineering query.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		
1	2026-10-27	SRS and plot plan issued; EQ-la queries added; EQ-ej-1, EQ-er-10 (with EQ-la-2/-3/-8) closed, EQ-ej-2 partly closed	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Document register

One row per document. Pages and paper size are read from each PDF (pdfinfo) when the register is built. "Built from" lists the documents whose numbers it takes; "Feeds" is the inverse. The instructor-only training material is registered in the instructor edition of the OTS page, not in this document.

Document no.	Title	Discipline	Rev	Status	Pages	Paper	Source tab	Built from	Feeds
CHX-V1-PM-DOC-001	Document register, engineering queries and open recommendations — this document	Project	1	Issued for teaching	26	A4 landscape	Project & deliverables	all documents	—
CHX-V1-PFD-001	Process flow diagram — also .svg, .png; editable .pptx in the zips	Process	0	Issued for teaching	1	A4 landscape	PFD	model	PE-HMB-001, PE-LL-001, PID-001-1, PID-001-2, ME-EQL-001, EE-HAC-001, SA-HZP-001
CHX-V1-PE-HMB-001	Heat and material balance (design + 120 % + 70 %)	Process	0	Issued for teaching	16	A4 landscape	Process engineering	PFD-001	PE-LL-001, PL-PP-001, ME-EQL-001, ME-DS-001, IN-INL-001, EE-LL-001, EE-HAC-001, PC-PKG-001
CHX-V1-PE-LL-001	Line list and line sizing	Process	0	Issued for teaching	12	A4 landscape	Process engineering	PE-HMB-001, PFD-001	PE-RLB-001, PL-PP-001, ME-EQL-001, ME-DS-001, IN-INL-001, PC-PKG-001, CM-PLN-001
CHX-V1-PE-RLB-001	Relief load basis (PSV/TSV sizing, flare, depressuring) — rev 1: elevations and relief line lengths from the plot plan (EQ-er-10)	Process	1	Issued for teaching	23	A4 landscape	Relief & vent	PE-LL-001, PID-001-1, PID-001-2, ME-EQL-001, SA-HZP-001, PL-PP-001	PL-PP-001, PC-PKG-001
CHX-V1-PID-001-1	EPC P&ID sheet 1 · feed, R-101, pump-around, E-101/D-101, E-103/V-103 — A3; also .svg, .png; editable .pptx in the zips	Piping	0	Issued for teaching	1	A3 landscape	P&ID	PFD-001	PE-RLB-001, PL-PP-001, ME-EQL-001, ME-DS-001, IN-INL-001, IN-CE-001, EE-LL-001, EE-HAC-001, CM-PLN-001
CHX-V1-PID-001-2	EPC P&ID sheet 2 · R-102, F-102, R-103, E-106, V-104, K-101, make-up, purge — A3; also .svg, .png; editable .pptx in the zips	Piping	0	Issued for teaching	1	A3 landscape	P&ID	PFD-001	PE-RLB-001, PL-PP-001, ME-EQL-001, ME-DS-001, IN-INL-001, IN-CE-001, EE-LL-001, EE-HAC-001, CM-PLN-001
CHX-V1-PL-PP-001	Plot plan, elevations, routing and piping isometrics — A3; data layout.json (also feeds the Visual plant 3D)	Piping	0	Issued for teaching	28	A3 landscape	Layout & isometrics	ME-EQL-001, PE-LL-001, PE-HMB-001, EE-HAC-001, PE-RLB-001, PID-001-1, PID-001-2, IN-INL-001	PE-RLB-001
CHX-V1-ME-EQL-001	Equipment list	Mechanical	0	Issued for teaching	15	A4 landscape	Mechanical	PFD-001, PE-HMB-001, PE-LL-001, PID-001-1, PID-001-2, EE-LL-001	PE-RLB-001, PL-PP-001, ME-DS-001, PC-PKG-001
CHX-V1-ME-DS-001	Equipment datasheet book (sheets CHX-V1-ME-DS-<tag>, R-101 GA)	Mechanical	0	Issued for teaching	20	A4 portrait	Mechanical	ME-EQL-001, PE-HMB-001, PE-LL-001, PID-001-1, PID-001-2	PC-PKG-001

Document no.	Title	Discipline	Rev	Status	Pages	Paper	Source tab	Built from	Feeds
CHX-V1-IN-INL-001	Instrument list, I/O count, control-valve and flow-element sizing	Instrumentation	0	Issued for teaching	28	A4 landscape	Instrumentation & control	PID-001-1, PID-001-2, IN-CE-001, PE-HMB-001, PE-LL-001	PL-PP-001, PC-PKG-001
CHX-V1-IN-CE-001	Cause-and-effect matrix (SIS trips, alarms, permissives)	Instrumentation	0	Issued for teaching	2	A4 portrait	C&E matrix	PID-001-1, PID-001-2, SA-HZP-001	IN-INL-001, IN-SRS-001, PC-PKG-001, CM-PLN-001
CHX-V1-IN-SRS-001	Safety requirements specification of the SIFs — same data as the SRS tab: sf.json (C&E), model process-safety-time runs, LOPA block diagrams	Instrumentation	0	Issued for teaching	29	A4 landscape	SRS of SIF	IN-CE-001, SA-HZP-001	PC-PKG-001, CM-PLN-001
CHX-V1-EE-LL-001	Electrical load list (motors, other loads, UPS, feeders, starting)	Electrical	0	Issued for teaching	12	A4 landscape	Electrical	PE-HMB-001, PID-001-1, PID-001-2	ME-EQL-001, PC-PKG-001
CHX-V1-EE-HAC-001	Hazardous-area classification schedule	Electrical	0	Issued for teaching	8	A4 landscape	Electrical	PFD-001, PE-HMB-001, PID-001-1, PID-001-2	PL-PP-001, PC-PKG-001
CHX-V1-SA-HZP-001	HAZOP report — study made on the screen P&ID (rev 0 of the OTS drawing)	Safety	0	Issued for teaching	72	A4 portrait	HAZOP	PFD-001	PE-RLB-001, IN-CE-001, IN-SRS-001, CM-PLN-001
CHX-V1-PC-PKG-001	Procurement packages (scope, key data, FAT/SAT, schedule)	Project	0	Issued for teaching	21	A4 landscape	Procurement to commissioning	ME-EQL-001, ME-DS-001, IN-INL-001, PE-RLB-001, EE-LL-001, EE-HAC-001, IN-CE-001, IN-SRS-001, PE-HMB-001, PE-LL-001	CM-PLN-001
CHX-V1-CM-PLN-001	Pre-commissioning and commissioning plan (test packs, SIS tests, PSSR, performance test)	Project	0	Issued for teaching	19	A4 landscape	Procurement to commissioning	PC-PKG-001, IN-CE-001, IN-SRS-001, PID-001-1, PID-001-2, PE-LL-001, SA-HZP-001	—

Studies (tab only) and deliverables not yet issued

Studies of this OTS feed the documents without a document number of their own. The deliverables in the second table are relied on by the documents but have not been issued (EQ-ej-2).

Study (tab)	Discipline	Feeds
Design of the unit operation	Process	PFD-001, ME-EQL-001
BFD	Process	PFD-001
Operating windows	Process	PE-HMB-001, EE-LL-001
HAZID	Safety	PL-PP-001, EE-HAC-001, SA-HZP-001
HAZOP worksheet	Safety	SA-HZP-001
Safety layers	Safety	PE-LL-001, PE-RLB-001, PID-001-1, PID-001-2, PL-PP-001, ME-EQL-001, IN-INL-001, IN-CE-001, IN-SRS-001, CM-PLN-001
LOPA	Safety	IN-SRS-001
Control loop design	Instrumentation	PID-001-1, PID-001-2, IN-INL-001
Process reliability	Electrical	EE-LL-001
SOP register	Project	CM-PLN-001

Study (tab)	Discipline	Feeds
Transition operation	Project	CM-PLN-001

Deliverable not issued	Discipline	Why it is needed
Piping material specification (classes A1 … B3)	Piping	the line list assigns classes; the package for bulk piping cites the specification
Single-line diagram	Electrical	bus sections for A/B motors and the UPS supply
Instrument and analyser datasheets	Instrumentation	packages PK-09, PK-11, PK-12 buy against them
Catalyst specification	Process	package PK-13 buys against it
Fire & gas cause-and-effect	Safety	detectors and deluge of the Safety-layers additions have no C&E

2 · Engineering-query register

How queries are handled: the discipline that finds a conflict raises a query with the documents involved and a proposed resolution; document control numbers it and assigns an owner and a priority; the owner answers; changes to a design basis or a safeguard are approved (management of change after the HAZOP); the documents are revised and the query is closed with the revision reference. IDs are those of the tab that raised each query; EQ-ej are this register's cross-document findings. Priority: **Safety** — a safeguard, relief case or design pressure/temperature is in question; **Design** — a size, rating or design value changes; **Documentation** — the documents disagree, the plant would not change. Status: 75 OPEN, 1 PARTLY CLOSED, 5 CLOSED; a closed query keeps its row with the closure reference.

Raised by	Safety	Design	Documentation	Total
Process engineering	5	6	2	13
Mechanical	4	4	3	11
Relief & vent	5	2	3	10
Instrumentation & control	2	4	3	9
Electrical	2	3	2	7
Procurement to commissioning	3	2	2	7
EPC P&ID	1	1	3	5
Layout & isometrics	3	2	4	9
Project & deliverables	4	1	5	10
All	29	25	27	81

Owner discipline: Project 3 · Process 45 · Safety 6 · Mechanical 7 · Piping 6 · Instrumentation 12 · Electrical 2

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-ep-1	Documentation	Process engineering / Process	**PFD has no stream for the FV-104 by-pass and the D-101 blowdown** P-101 always moves 606 t/h; at design 248 t/h of it by-passes E-101 through the three-way FV-104 (stream 10B here), and D-101 blows down 1 % of the steam (BD). PFD stream 10 shows only the 357 t/h through E-101, so the PFD alone under-states the pump and the return line by 70 %.	CHX-V1-PFD-001, CHX-V1-PE-HMB-001, P-101 datasheet	Add 10B and BD to PFD rev 1; P-101 and lines 16"-P-1004-B3, 16"-P-1006-B3 rated at the full pump flow.	**OPEN**	em-6 er-8 px-3 px-5 la-5 ej-8
EQ-ep-2	Design	Process engineering / Process	**P-101 differential pressure has two values** The page's streamRows puts the pump-around discharge at PI-101 + 2 bar; the Flexibility tab illustration uses 3 bar at 606 t/h (shut-off 4.0). The hydraulic budget in part 6 needs the lines + E-101 + FV-104 to fit inside the rated differential.	ui2.js streamRows, Flexibility tab, CHX-V1-PE-LL-001, P-101 datasheet	Fix the rated differential on the P-101 datasheet from part 6 (E-101 allowable ΔP set from the budget); align streamRows and the Flexibility illustration (lead).	**OPEN**	em-9 er-2 er-3 ei-2 ei-3
EQ-ep-3	Safety	Process engineering / Process	**MP steam reaches the E-101 shell protected at 8 bar g** HS-101 HEAT puts MP steam (10 bar g) on the E-101 shell, which is the D-101 water side protected by PSV-109 at 8 bar g (SL_ASSUMED). Line 8"-S-3002-A1 is designed for 12 bar g but the equipment it feeds is not.	CHX-V1-PE-LL-001, Safety layers SL-R12, CHX-V1-PE-RLB-001, D-101/E-101 datasheets	Either D-101/E-101 shell design ≥ 12 bar g with PSV-109 re-set, or a pressure-reducing station + PSV on 8"-S-3002-A1 (Relief & vent tab).	**OPEN**	em-2 er-4 ec-2 px-2 ej-9

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-ep-4	Safety	Process engineering / Process	**R-102 jacket: steam and cooling water on the same jacket** The jacket takes MP steam (10 bar g, 184 °C) in HEAT and CW in COOL. TSV-102J is set at 10 bar g = the MP steam pressure (relief at operating pressure), and the CW lines up to the change-over valves see steam.	Safety layers SL-R15, CHX-V1-PE-LL-001 6"-CW-3011-A1, R-102 datasheet	Jacket and CW branch up to the block valves designed 12 bar g / 200 °C; TSV-102J set ≥ 12 bar g; double block and bleed at the HS-102 change-over.	**OPEN**	ep-5 em-11 ei-1
EQ-ep-5	Design	Process engineering / Process	**TSV-102F set close to the F-102 operating pressure** F-102A/B sit on the P-102 discharge (stream 14 32.6 bar g at design, 32.9 at 120 %). TSV-102F is set at the loop design 36 bar g, so the operating pressure is 91 % of the set (typical limit 90 % for spring valves).	Safety layers SL-R16, CHX-V1-PE-LL-001 3"-P-1011-B3, F-102 datasheet	F-102 design 40 bar g (= the pump discharge side, as 3"-P-1011-B3) and TSV-102F set 40 bar g.	**OPEN**	ep-4 em-11 ei-1
EQ-ep-6	Safety	Process engineering / Process	**Nitrogen flow in the model is far above a plant N_2 supply** HV-120 delivers 0.35·√(9 − P) kmol/s: 79,814 Nm³/h into an empty loop, which makes 16"-N-3012-A1. A typical purge supply is a few thousand Nm³/h (assumed), and nothing stops loop gas flowing back into the 8 bar g header.	model.js (Qn2), CHX-V1-PE-LL-001, P&ID CHX-V1-PID-001	Define the N_2 header capacity, add a restriction orifice RO-120 + check valve and double block at the tie-in; cap the model N_2 flow (lead, model change).	**OPEN**	
EQ-ep-7	Design	Process engineering / Process	**Cooling-water return temperatures above typical practice** E-103 returns 47.7 °C at K.Fcw3 = 120 kg/s and E-106 45.9 °C at the econ() 60 kg/s (design); at 120 % E-106 returns 48.7 °C. Typical maximum return 45 °C (fouling, scaling). The model has no E-106 water flow.	model.js K.Fcw3, econ(), CHX-V1-PE-HMB-001, E-103/E-106 datasheets	E-103 CW ≥ 145 kg/s and E-106 ≥ 77 kg/s for a 45 °C return; put the E-106 flow on its datasheet.	**OPEN**	ej-4
EQ-ep-8	Design	Process engineering / Process	**Recycle gas carries no cyclohexane vapour in the model** At V-104 (44.9 °C, 30.0 bar a) the model's own PsatC gives 1.1 mol % cyclohexane in the gas ≈ 1.84 t/h circulating through K-101 and 81 kg/h lost to fuel gas; the HMB follows the model (dry gas).	model.js, CHX-V1-PE-HMB-001, K-101 datasheet	K-101 datasheet with the saturated gas (molar mass and flow); product loss to fuel gas in the economics.	**OPEN**	la-6
EQ-ep-9	Design	Process engineering / Instrumentation	**FV-103 at 92 % open in the 120 % case** The design-basis maximum rate needs FV-103 at 92.1 % (design 50 %, 70 % case 27 %); typical practice keeps a control valve ≤ 80–90 % open at maximum flow. The Operating-windows tab finds FV-103 as the top of the window.	CHX-V1-PE-HMB-001, Operating windows / Flexibility tabs, FV-103 datasheet	Size FV-103 for the 120 % case at ≤ 80 % (larger Cv) together with K-101 head (Flexibility recommendation).	**OPEN**	
EQ-ep-10	Documentation	Process engineering / Process	**Power in econ() differs from the hydraulic estimates** econ() books K-101 1200 kW, P-101 160 kW, P-100 90 kW, P-102 30 kW; part 6 estimates the absorbed power from flow × head (see the table).	model.js econ(), CHX-V1-EE-LL-001	Electrical load list from the equipment datasheets (absorbed power + motor margin); keep econ() as a training price model and say so.	**OPEN**	ee-1 ee-2 ee-3 ee-5
EQ-ep-11	Safety	Process engineering / Mechanical	**Hot-hydrogen material for spec B2** B2 lines run at up to 200 °C (design 250–260 °C) with a hydrogen partial pressure ≈ 14 bar. Carbon steel in hot hydrogen needs an API 941 (Nelson curve) check.	CHX-V1-PE-LL-001 spec B2, CHX-V1-ME-EQL-001	Mechanical to confirm B2 material (killed CS + PWHT or Cr-Mo) against API 941 at the design temperature.	**OPEN**	

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-ep-12	Safety	Process engineering / Process	**Product line downstream of LV-104 and V-104 design temperature** 4"-P-1014-A2 (class 150, 10 bar g assumed) sees loop gas at 36 bar g if V-104 loses its liquid seal (Safety layers SL-R6 recommends LSLL-104 + XV-104). On loss of E-106 cooling the V-104 inlet reaches the R-103 temperature (159 °C) while its outlets are designed for operating + 30 °C.	CHX-V1-PE-LL-001, Safety layers SL-R6, CHX-V1-PE-RLB-001, V-104 datasheet	Gas blow-by relief case on the product line/tanks (Relief & vent tab) or spec B1 up to a second isolation; V-104 design temperature to cover loss of E-106 cooling.	**OPEN**	em-10 ei-8
EQ-ep-13	Design	Process engineering / Process	**Catalyst slurry below 1 m/s at the lowest case** 16"-P-1005-B3 0.90 m/s, 16"-P-1007-B3 0.61 m/s, 4"-P-1010-B3 0.77 m/s: the E-101 branch and the FV-104 by-pass swap flow as FV-104 moves, and the P-102 suction cannot hold 1–2 m/s over a 1.7 : 1 flow range with standard sizes (3" gives 2.30 m/s at the rated flow).	CHX-V1-PE-LL-001, P&ID CHX-V1-PID-001, P-101/P-102 datasheets	Flushing connections (cyclohexane) on the three lines; FV-104 minimum by-pass stop; accept 3" P-102 suction at the rated case or a minimum-flow recycle on P-102; settle the catalyst loading (not in the model).	**OPEN**	
EQ-em-1	Safety	Mechanical / Safety	**R-101 design temperature below the boiling ceiling** The R-101 design temperature 250 °C (SL_ASSUMED.Tdes1, also the R-101 line design temperature) is below the temperature the reactor can reach if the trips fail: the boiling ceiling TbC(PI-101) = 258.3 °C, 272.4 °C at the PSV set. SA-516 Gr 70 keeps S = 138 MPa up to 343 °C, so raising the design temperature costs no thickness.	CHX-V1-ME-EQL-001, Safety layers (SL_ASSUMED.Tdes1, R-101 ladder), CHX-V1-PE-LL-001 (R-101 lines 250 °C)	R-101, E-101 tubes, P-101, the R-101 lines and D-102 designed for 280 °C (≥ the ceiling at the PSV set); Safety layers to keep 250 °C as the critical limit of the ladder or move it.	**OPEN**	er-1 er-6 ee-4
EQ-em-2	Safety	Mechanical / Process	**E-101 shell: two pressure sources, one PSV** In COOL the E-101 shell is the D-101 water side (PSV-109 8 bar g). In HEAT HY-101A/B isolate it and MP steam (10 bar g, line design 12 bar g) feeds it, so PSV-109 does not protect it. This EQL designs the E-101 shell for 12 bar g / 200 °C and keeps D-101 at 8 bar g. Failure of HY-101A/B to close puts MP steam into D-101.	CHX-V1-ME-EQL-001, CHX-V1-PE-LL-001 8"-S-3002-A1, Safety layers SL-R12, EQ-ep-3, P&ID sheet 1	Relief & vent: fire case on the isolated E-101 shell (own PSV or thermal relief) and the HY-101A/B-fail case on PSV-109 (MP steam through HY-101D); or D-101 also 12 bar g with PSV-109 re-set.	**OPEN**	ep-3 er-4 ec-2 px-2 ej-9
EQ-em-3	Safety	Mechanical / Process	**Tube rupture is a relief case on E-101, E-103 and E-106** The low-pressure side is below 10/13 of the high-pressure side design: E-101 shell 12 vs tubes 40 bar g (slurry and dissolved H_2 into D-101); E-103 and E-106 CW tubes 10 vs shell 36 bar g (hydrogen into the cooling-water system). TSV-113/TSV-106 are thermal reliefs only.	CHX-V1-ME-EQL-001 (E-101/E-103/E-106 sheets), Safety layers SL_IT, Relief & vent (CHX-V1-PE-RLB-001), HAZOP	Relief & vent to size tube-rupture relief (or raise the low-pressure side to 10/13 of the other: E-101 shell 31 bar g, CW sides 28 bar g); CW return gas detection (HAZOP).	**OPEN**	er-5
ID	**Priority**	**Raised by / owner**	**Query**	**Documents involved**	**Proposed resolution**	**Status**	**Same topic**
EQ-em-4	Documentation	Mechanical / Piping	**Nozzle sizes: line list and EPC P&ID differ** Nozzles are sized to the line list CHX-V1-PE-LL-001. The EPC P&ID (its own numbering SIZE-SERVICE-NNNN) draws 26 of these connections at another size: R-101 N1 3"-P-1002-B1 vs P&ID 4"-P-1002; R-101 N4 8"-P-1009-B1 vs P&ID 6"-P-1007; R-101 N7 4"-P-1008-B3 vs P&ID 6"-SL-1014; R-102 N1 4"-P-1008-B3 vs P&ID 6"-SL-1014;	CHX-V1-PID-001 sheets 1–2, CHX-V1-PE-LL-001, all datasheets	P&ID rev 1 to adopt the line-list numbers and sizes (one numbering in the unit); nozzles stay as the line list.	**OPEN**	em-5 er-9 ei-6 la-4 ej-3

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
			R-102 N4 4"-P-1010-B3 vs P&ID 6"-SL-2001; R-102 J1 6"-S-3003-A1 vs P&ID 4"-MS-2020; R-103 N3 8"-PG-2005-B2 vs P&ID 10"-P-2005; E-101 N1 16"-P-1005-B3 vs P&ID 12"-SL-1015; E-101 N2 16"-P-1006-B3 vs P&ID 12"-SL-1012; E-101 N3 8"-BFW-3005-A1 vs P&ID 12"-BW-1022; E-101 N4 16"-S-3006-A1 vs P&ID 14"-LS-1023; E-103 N2 8"-PG-2002-B2 vs P&ID 6"-P-1006; E-106 N1 8"-PG-2005-B2 vs P&ID 10"-P-2005; E-106 N3 8"-CW-3009-A1 vs P&ID 10"-CW-2030; E-106 N4 8"-CW-3010-A1 vs P&ID 10"-CW-2030; D-101 N3 8"-BFW-3005-A1 vs P&ID 12"-BW-1022; D-101 N4 16"-S-3006-A1 vs P&ID 14"-LS-1023; V-103 N1 8"-PG-2002-B2 vs P&ID 6"-P-1006; V-103 N3 8"-P-1009-B1 vs P&ID 6"-P-1007; V-104 N2 8"-H-2007-B2 vs P&ID 10"-G-2007; V-104 N3 4"-P-1013-B1 vs P&ID 6"-P-2011; V-104 N4 1-1/2"-H-2011-B2 vs P&ID 2"-G-2010; V-104 N5 6"-H-2010-B1 vs P&ID 8"-G-2009; P-100 D 3"-P-1002-B1 vs P&ID 4"-P-1002; P-102 S 4"-P-1010-B3 vs P&ID 6"-SL-2001; K-101 S 8"-H-2007-B2 vs P&ID 10"-G-2007.				
EQ-em-5	Documentation	Mechanical / Process	**Nozzle connections without a line-list line** 11 nozzles connect to lines drawn on the P&ID but missing from the line list: R-101 N8 (4"-SL-1004); R-101 N9 (6"-P-1003); R-101 N10 (6"-FL-1040); R-102 N5 (4"-FL-2015); R-102 N6 (8"-FL-2016); R-102 J3 (6"-CW-2022); R-102 J4 (2"-CD-2023); E-101 N6 (3"-CD-1025); D-101 N5 (1"-BD-1026); D-102 N1 (4"-SL-1004); V-104 N6 (8"-FL-2016).	CHX-V1-PE-LL-001, CHX-V1-PID-001, datasheets	Add these lines (catalyst charge, start-up solvent, relief inlets, jacket return and condensate, E-101 condensate, blowdown, R-102 off-gas upstream of EDV-101) to the line list rev 1.	**OPEN**	em-4 er-9 ei-6 la-4 ej-3
EQ-em-6	Documentation	Mechanical / Process	**PFD equipment sizes differ from the model spans** PFD sizes: R-101 Ø3.6 × 4.5 m (Design tab 3.80 × 4.50); V-104 vertical Ø2 × 5 m ≈ 17.8 m³ against a model level span of 30 m³; D-101 Ø1.4 × 4 m ≈ 6.9 m³ against a water span of 24 t (24.9 m³). This EQL sizes from the model: V-104 2.30 × 9.1 m horizontal, D-101 2.40 × 8.4 m.	CHX-V1-PFD-001 (pfspec real sizes), Design tab, model K.V4span / K.Mspan, CHX-V1-ME-EQL-001	PFD rev 1 drawing sizes from the EQL; or reduce the model spans to the PFD vessels (lead: model change, level dynamics change).	**OPEN**	ep-1 er-8 px-3 px-5 la-5 ej-8
EQ-em-7	Design	Mechanical / Process	**M-102 agitator on the P&ID, not in the model or the load list** R-102 is a slurry reactor and the P&ID draws M-102, but the model has no R-102 agitator (MOTORS) and the load list has no M-102. Kept on the EQL as HOLD with ≈ 23 kW (1.5 kW/m³ typical, assumed) → 30 kW.	CHX-V1-ME-EQL-001, CHX-V1-EE-LL-001, CHX-V1-PID-001 sheet 2, EQ-ec-1, EQ-px-4	Process to confirm R-102 mixing; if agitated, add M-102 to the load list and the C&E (stop on trip?) and to the model MOTORS (lead).	**OPEN**	ec-1 px-4
EQ-em-8	Safety	Mechanical / Process	**D-102 catalyst charging has no pressure source** D-102 is filled at ambient pressure under N_2 (8 bar g header) but discharges into R-101 at 30 bar g through HV-117. The design pressure here is the loop 36 bar g because D-102 sees R-101 when HV-117 is open.	CHX-V1-PID-001 sheet 1, CHX-V1-ME-DS-D-102, SOP (catalyst make-up)	Define charging: at start-up below the N_2 pressure only, or pressurise D-102 from the K-101 discharge (then design 42 bar g), or a slurry pump; add the procedure to the SOP.	**OPEN**	ec-3 ec-5 ej-6

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-em-9	Design	Mechanical / Process	**Model heat-transfer capacities above the datasheet equipment** E-103: model UA_3 500 kW/K (capacity 47.9 MW at design CW) vs 163 kW/K for the datasheet area at U 0.45 (assumed). R-102 jacket: model UAj 90 kW/K on 26.4 m² of wetted jacket needs U 3.41 kW/m²K (typical jacket 0.3–0.6).	model.js K.UA3, K.UAj; Flexibility tab (capacities); CHX-V1-ME-DS-E-103, -R-102	Either R-102 gets an internal coil (≈ 180 m² at U 0.5) and E-103 the larger area, or the model UA values are reduced to the datasheets (lead: model change; the Flexibility capacities fall).	**OPEN**	ep-2 er-2 er-3 ei-2 ei-3
EQ-em-10	Design	Mechanical / Process	**V-104 design temperature vs its outlet lines** V-104 is designed for 190 °C here (loss of E-106 cooling: R-103 outlet 158.6 °C + 30 °C, as EQ-ep-12 asks). The line list keeps the V-104 outlets (8"-H-2007-B2, 4"-P-1013-B1, 1-1/2"-H-2011-B2) at 80 °C and the inlet 8"-PG-2006-B2 at 260 °C.	CHX-V1-ME-DS-V-104, CHX-V1-PE-LL-001	Line list rev 1: V-104 outlet lines to the vessel design temperature, or a TSHH on the E-106 outlet that trips the unit (Safety layers).	**OPEN**	ep-12 ei-8
EQ-em-11	Design	Mechanical / Process	**Relief set pressures to follow the equipment design pressures** F-102A/B are designed 40 bar g here (P-102 discharge side, EQ-ep-5) but TSV-102F is set at 36 bar g; the R-102 jacket is 12 bar g but TSV-102J is 10 bar g = the MP steam pressure (EQ-ep-4).	Safety layers SL-R15/SL-R16, CHX-V1-ME-DS-F-102, -R-102, Relief & vent	TSV-102F set 40 bar g and TSV-102J 12 bar g (Safety layers / Relief & vent).	**OPEN**	ep-4 ep-5 ei-1
EQ-er-1	Safety	Relief & vent / Process	**Loop relief sits at the critical point of cyclohexane** With the loop PSVs set at 36 bar g the relieving pressure is 40.61 bar a (10 %) and 44.57 bar a (fire, 21 %); the model's own correlations put the cyclohexane critical point at 280.4 °C / 40.7 bar a (PsatC at the Watson Tc of lamC). A blocked-in vessel in a fire is therefore supercritical and the fire loads use the 115 kJ/kg floor (typical practice) — uncertain. The cooling-water case stays below the critical point only because the recycle gas keeps sweeping R-101 (relieving at 255.5 °C); it needs Q for 178.7 t/h and sets the unit flare header.	CHX-V1-PE-RLB-001, SL_ASSUMED (Safety layers), CHX-V1-ME-EQL-001	Do a dynamic (supercritical-fluid) fire relief study; evaluate an SIS trip of feed and K-101 on loss of CW (HAZOP R06) as a high-integrity alternative under API 521, which would take the cooling-water case off the flare.	**OPEN**	em-1 er-6 ee-4
EQ-er-2	Documentation	Relief & vent / Process	**The model holds no vapour or liquid volume in the loop pressure** Loop pressure is Ps = nG·RTV (permanent gas only, fixed 80 m³). Boil-up that is not refluxed becomes R-102 liquid (cooling-water run: vapour not refluxed 54 t/h, PI-101 unchanged), and a rising V-104 level does not compress the gas (blocked-outlet run: level 110 %, PI-105 unchanged). The model cannot show the over-pressure of these cases.	model.js step(), CHX-V1-PE-RLB-001 runs cw, lv104	Keep the hand calculation as the basis; for teaching, add a condensable-vapour term to the loop inventory and subtract the liquid volumes from K.Vloop (lead / model owner).	**OPEN**	ep-2 em-9 er-3 ei-2 ei-3
EQ-er-3	Design	Relief & vent / Process	**Model caps the make-up at muMax** The make-up flow is capped at K.muMax = 0.9 kmol/s (72621 Nm³/h) whatever PV-105 and the header pressure allow. With the header at 45 bar g (assumed) the cap is active when PI-101 crosses the set (run pv105hi), so the run's rise rate is a model limit. The RLB load uses the valve equation at the relieving pressure (0.752 kmol/s, below the cap).	model.js K.muMax, SRS PST run, CHX-V1-PE-RLB-001	Confirm the maximum make-up header pressure (supplier) and the PV-105 Cv on its datasheet; size PSV-104 on that pressure.	**OPEN**	ep-2 em-9 er-2 ei-2 ei-3

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-er-4	Safety	Relief & vent / Process	**E-101 shell in HEAT: MP steam 10 bar g against an 8 bar g relief (and no shell pressure in the model)** HS-101 HEAT puts MP steam (10 bar g) on the E-101 shell. The equipment list designs the E-101 shell for 12 bar g but D-101 for 8 bar g: if the shell stays open to D-101, PSV-109 (set 8 bar g) must pass the full HV-110 flow (15.1 t/h) — this case governs PSV-109 (P). If the shell is isolated from D-101 in HEAT, the shell has no relief device at all. The model takes D-101 off E-101 in HEAT and has no shell pressure, so the run cannot show it.	EQ-ep-3, SL-R12, CHX-V1-PE-LL-001 8"-S-3002-A1, CHX-V1-ME-EQL-001 E-101 / D-101	Design D-101 and its circuit for ≥ 12 bar g with PSV-109 re-set, or put a pressure-reducing valve plus a PSV on 8"-S-3002-A1; give the E-101 shell its own PSV if it can be isolated from D-101; interlock HEAT with XV-101 (R16).	**OPEN**	ep-3 em-2 ec-2 px-2 ej-9
EQ-er-5	Safety	Relief & vent / Process	**Exchanger tube rupture is credible on E-101, E-103 and E-106** Low-pressure sides (E-101 shell 8, CW sides 10 bar g) are below 10/13 of the tube-side design pressure (40 / 36 bar g). E-101: 78.9 t/h of hot slurry flashing (16 % vapour) into the steam system, vented to atmosphere through PSV-109. E-103 / E-106: loop gas into the cooling water that the thermal relief valves cannot pass.	CHX-V1-PE-RLB-001, CHX-V1-ME-EQL-001, HAZOP R15, R17	Raise the low-pressure-side design pressure to ≥ 10/13 of the high side, or add rupture-disc relief on the CW returns to a safe location and route PSV-109 away from grade; detect hydrocarbon in the D-101 blowdown and CW returns (R15, R17).	**OPEN**	em-3
EQ-er-6	Safety	Relief & vent / Safety	**Runaway relief is two-phase and outside this basis** Loss of both heat sinks with hydrogen at FV-103 full open releases 32.7 MW; the vapour-only lower bound is 762 t/h (2 × T), far above the selected Q. The real relief of a foaming slurry with dissolved hydrogen is two-phase; DIERS methods are outside the scope.	CHX-V1-PE-RLB-001, LOPA L1, SRS SIF-1	Treat the double failure through the SIS (TSHH-100, recommended LSLL/TDA items) and LOPA L1; if it stays a relief case, commission a DIERS two-phase study.	**OPEN**	em-1 er-1 ee-4
EQ-er-7	Safety	Relief & vent / Process	**Liquid flash dominates EDV-101 depressuring; model EDV ignores it** To reach 6.9 bar g in 15 min the loop gas alone needs 172 mm²; the R-101 liquid flashing below Psat(200 °C) = 12.4 bar g adds 59.8 kmol of cyclohexane and raises the orifice to 1756 mm². The model's EDV-101 (equivalent 1087 mm², gas only) reaches 6.9 bar g in 130 s; with the flash counted the same orifice is still at 8.4 bar g after 15 min. The isentropic gas temperature falls to -54.6 °C (cold lower bound).	HAZOP R29, model.js Qvent, CHX-V1-PE-LL-001 8"-FL-3013-A2	Size the EDV-101 restriction orifice on the flash basis; check minimum design metal temperature of the loop (R29) and the flare header for the cold gas.	**OPEN**	ei-4 ee-6
EQ-er-8	Documentation	Relief & vent / Process	**V-104 size: PFD drawing smaller than the model level span** The model's V-104 level span is 30 m³ (50 % = 15 m³ of liquid) while the PFD vessel 2.0 × 5.0 m holds 17.8 m³. The equipment list sizes V-104 2.3 × 9.1 m; this RLB uses that size for the fire case (wetted 33.4 m²).	model.js K.V4span, CHX-V1-PFD-001, CHX-V1-ME-EQL-001	Revise the PFD drawing size of V-104 to the equipment list; keep K.V4span consistent.	**OPEN**	ep-1 em-6 px-3 px-5 la-5 ej-8
ID	**Priority**	**Raised by / owner**	**Query**	**Documents involved**	**Proposed resolution**	**Status**	**Same topic**
EQ-er-9	Design	Relief & vent / Piping	**Tailpipe sizes on the P&ID (HOLD) against the selected valves** PSV-101: valve outlet 8", P&ID 6", RLB 24" (built-up + superimposed 1.1 % of set, exit Mach 0.5); PSV-102:	CHX-V1-PID-001 6"-FL-1040, 8"-FL-2016 (HOLD), CHX-V1-PE-RLB-001	Release the HOLDs with the sizes of part 5 (one tailpipe per valve; the shared 8"-FL-2016 becomes a sub-header), and add the unit flare	**OPEN**	em-4 em-5 ei-6 la-4 ej-3

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
			valve outlet 4", P&ID 8", RLB 10" (built-up + superimposed 1.3 % of set, exit Mach 0.47); PSV-104: valve outlet 4", P&ID 8", RLB 8" (built-up + superimposed 1.6 % of set, exit Mach 0.45); PSV-112: valve outlet 4", P&ID 8", RLB 8" (built-up + superimposed 1.1 % of set, exit Mach 0.52). The tailpipes were enlarged until the total back pressure is ≤ 10 % of set and the exit Mach ≤ 0.7 (typical practice); PSV-101 needs a 24" tailpipe because of the 179 t/h cooling-water case (EQ-er-1).		header to the line list.		
EQ-er-10	Documentation	Relief & vent / Piping	**Equipment elevations and relief line lengths: now from the plot plan (closed)** The fire cases depend on the height of each vessel above grade (7.6 m rule). This revision reads the plot plan CHX-V1-PL-PP-001 (tabs/lay/layout.json): bottoms at R-101 7.95 m (was 4), R-102 5.8 m (was 2), R-103 2.49 m (was 1.5), V-103 15.3 m (was 9), V-104 1.45 m (horizontal drum as on the EQL and the plot plan; was vertical at 1), D-101 8.4 m, E-101 shell 0.98 m. Fire loads: PSV-101 (R-101) 12.9 t/h, PSV-102 (R-102 + R-103) 39.8 t/h, PSV-104 (V-104) 24 t/h. Lengths: tailpipes PSV-101 20.5 m, PSV-102 17.8 m, PSV-104 19 m, PSV-112 6.4 m (were 20 m); inlets PSV-104 5.3 m and PSV-112 6.1 m (were 1.5 m); unit header to V-190 176 m (was 150 m); PSV-109 vent stack 9.9 m (was 4 m). Result: orifices PSV-101 Q, PSV-102 K, PSV-104 J, PSV-112 J, PSV-109 P; inlet losses ≤ 2.9 % of set; total back pressure ≤ 4.2 % of set; unit header 30" (inlet 1.312 bar a, Mach 0.26). Still assumed: the F-102 projected area for solar heating.	CHX-V1-PL-PP-001 (EQ-la-2, EQ-la-3, EQ-la-8), CHX-V1-ME-EQL-001, CHX-V1-PE-RLB-001	Closed in this revision of the RLB (layout.json elevations and lengths). The plot plan copy of the RLB (layout.json src.rlb) refreshes when the Layout tab re-runs la_data.js; F-102 projected area stays assumed.	**CLOSED** Closed by CHX-V1-PE-RLB-001 rev 1 (2026-10-27): elevations and relief line lengths read from the plot plan CHX-V1-PL-PP-001 (layout.json).	la-2 la-9
EQ-ei-1	Design	Instrumentation & control / Instrumentation	**TV-102 cannot serve both MP steam and cooling water** Sized on MP steam, TV-102 needs Cv 27.9 (rated 69.3); in COOL the same valve must pass the model's cooling-water flow K.Fcwj = 40 kg/s, which needs Cv 118 at 2 bar (assumed) — 1.7 × the steam valve. The model lets one valve pass either fluid at its own capacity (steam ≤ K.msj · x, water K.Fcwj · x).	model.js (TV-102 jacket), EPC P&ID sheet 2 (TV-102 + HY-102A/B), CHX-V1-IN-INL-001	Two valves on split range from TIC-102: TV-102A steam (Cv 69.3) and TV-102B cooling water (Cv ≈ 120); or keep HY-102A/B as route valves and give each route its own control valve. Model: no change needed for training; note it on the P&ID.	**OPEN**	ep-4 ep-5 em-11
EQ-ei-2	Documentation	Instrumentation & control / Instrumentation	**Model valve capacities differ from the sized valves** Equivalent Cv of the model constants at 100 % (same IEC equations) against the rated Cv selected here: FV-101 33 vs 69.3 (× 0.48); FV-103 177 vs 400 (× 0.44); HV-107 7.1 vs 3 (× 2.36). The model's valves are linear and sized to sit near 50 % at design; a real valve is chosen for the 120 % case at ≤ 85 % open with its own characteristic.	model.js K.Cv101…Cv110, FBmax, Fpa, msj; CHX-V1-IN-INL-001 part 4	Keep the model constants for training (they reproduce the HMB); the valve datasheets use the rated Cv of this list. Instructors read the DCS output as "% of the model capacity", not as stem travel.	**OPEN**	ep-2 em-9 er-2 er-3 ei-3

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-ei-3	Documentation	Instrumentation & control / Instrumentation	**Valve opening on the DCS is not the real stem position** The model's valve output is linear in flow; an equal-percentage valve of the rated Cv sits elsewhere at design: PV-105 model 50 % vs 77 %; TV-102 model 50 % vs 76 %; LV-109 model 50 % vs 67 %; PV-110 model 50 % vs 71 %; HV-107 model 25 % vs 59 %.	model.js step() (P.v), CHX-V1-IN-INL-001 part 4, SOP / Operating windows texts quoting openings	Quote valve openings in procedures as "% of capacity (model)"; on a real plant use the travel from the valve datasheet. No model change.	**OPEN**	ep-2 em-9 er-2 er-3 ei-2
EQ-ei-4	Safety	Instrumentation & control / Instrumentation	**Instrument-air failure moves only the 11 control valves in the model** On loss of air the model drives FV/PV/LV/TV/HV-107 to the FAIL map; XV-101, XV-105, XV-104 (rec) close and EDV-101 opens on the P&ID (fail positions FC / FO assumed), and the HY route valves go to their FO/FC positions — the model keeps them where they were.	model.js step() FAIL map and air failure, EPC P&ID valve register, C&E CHX-V1-IN-CE-001	Confirm EDV-101 fail position (FO opens the loop to flare on any air loss: usually FO with a lock-in volume tank, or FC with SIS energise-to-open); add XV and HY valves to the model's air-failure response (lead, model.js).	**OPEN**	er-7 ee-6
EQ-ei-5	Design	Instrumentation & control / Instrumentation	**R-103 bed temperature range starts at 100 °C** TI-107A…E (and TSHH-107) read 100–250 °C (IND, ui1.js). A cold or start-up bed (≈ 30 °C) is below the range: the transmitters go under-range and the hottest-of-five logic sees a "bad PV".	ui1.js IND ranges, CHX-V1-IN-INL-001, SRS (TSHH-107)	Calibrate TT-107A…E 0–300 °C (normal 158.5 °C at 53 % of span; trip 220 °C inside); change IND r in ui1.js (lead).	**OPEN**	
EQ-ei-6	Documentation	Instrumentation & control / Piping	**Line sizes at the instruments differ between the EPC P&ID and the line list** FV-101 / FT-101: P&ID 4"-P-1002 vs line list 3"-P-1002-B1; FV-104 / FT-104 (E-101 leg): P&ID 12"-SL-1015 vs line list 16"-P-1005-B3; LV-101: P&ID 6"-SL-1014 vs line list 4"-P-1008-B3; PV-105 / FT-106: P&ID 8"-G-2009 vs line list 6"-H-2010-B1; HV-107 / FT-107: P&ID 2"-G-2010 vs line list 1-1/2"-H-2011-B2; LV-104 / FT-110: P&ID 6"-P-2011 vs line list 4"-P-1013-B1; TV-102 (steam): P&ID 4"-MS-2020 vs line list 6"-S-3003-A1. The P&ID used velocity limits of its own before CHX-V1-PE-LL-001 existed; this list and the valve sizing use the line list.	EPC P&ID CHX-V1-PID-001 (line labels), CHX-V1-PE-LL-001	Re-issue the P&ID line labels from CHX-V1-PE-LL-001 (numbers and sizes) at rev 1.	**OPEN**	em-4 em-5 er-9 la-4 ej-3
EQ-ei-7	Safety	Instrumentation & control / Instrumentation	**HV-113 permissive has no signal** The benzene start permissive "HV-113 open" (C&E, purgePermissives()) reads the position of a hand valve. A hand valve gives the SIS nothing to read.	C&E CHX-V1-IN-CE-001, model purgePermissives(), EPC P&ID sheet 1	Add limit switch ZSO-113 (SIS DI) on HV-113, or replace the permissive by a CW flow/temperature measurement on E-103 (HAZOP R06 style).	**OPEN**	ec-4 ej-7
EQ-ei-8	Design	Instrumentation & control / Instrumentation	**LV-104 letdown cavitates and releases dissolved hydrogen** LV-104 drops the product from 30.0 to 3.6 bar a (ΔP 26.4 bar). With a standard globe trim (FL 0.9) the choked limit is 24.0 bar, below the ΔP: the valve would choke and cavitate; the multi-stage anti-cavitation trim chosen here (FL 0.95, typical) raises the limit to 26.8 bar, and the cavitation index xF 0.89 is still above its incipient value. The liquid is also saturated with hydrogen at 29 bar g, so gas comes out of solution in the valve and the line to the tanks. The model treats LV-104 as a plain √ΔP valve.	model.js (L4), CHX-V1-PE-LL-001 4"-P-1014-A2, CHX-V1-IN-INL-001	Multi-stage anti-cavitation trim, hardened; outlet body and line sized for the released gas; keep the tank vent/flare route (EQ-ep-12).	**OPEN**	ep-12 em-10

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EQ-ei-9	Design	Instrumentation & control / Process	**FV-101 pressure drop rests on a flat pump curve** The model has no P-100 curve (feed = K.FBmax · x unless PI-101 ≥ 36 bar a), so the FV-101 ΔP here is the eproc estimate "shut-off − required" (4.53 bar at design). A real centrifugal pump loses head with flow and the valve ΔP at 120 % is smaller.	model.js (FB), CHX-V1-PE-LL-001 part 6, P-100 datasheet	Fix the P-100 rated point and curve on its datasheet and re-size FV-101 (Cv margin is large: 75 % at 120 %).	**OPEN**	la-1
EQ-ee-1	Design	Electrical / Process	**K-101 power in econ() is about 6 × a gas-compression estimate** K-101 power in econ() is 1200.2 kW at design, about 6.2 × a gas-compression estimate (194.1 kW for 46 000 Nm³/h at a pressure ratio 1.100, k 1.4, η 0.75 assumed). The load list carries the model value, which sets a 1600 kW MV motor and most of the demand.	model econ() · LL-001 · Process engineering (compressor hydraulics) · Mechanical (K-101 datasheet)	Process engineering to state the K-101 type, rated flow and head; if the absorbed power is near the estimate, K-101 needs about a 250 kW motor and the MV board, transformer and starting study shrink. Until then keep the conservative model value.	**OPEN**	ep-10 ee-2 ee-3 ee-5
EQ-ee-2	Documentation	Electrical / Electrical	**econ() charges shaft power as electricity** econ() charges the motors' absorbed (shaft) power as electricity (1633.2 kW); with typical motor efficiencies the electrical input is 1690.3 kW, plus 50.9 kW of other continuous loads.	model econ() · Efficiency tab (43.6 kWh/t) · LL-001	State in the Efficiency tab that its power is shaft power; or divide by η in econ() (lead decision: model change).	**OPEN**	ep-10 ee-1 ee-3 ee-5
EQ-ee-3	Documentation	Electrical / Process	**Pump powers in econ() are constants that do not follow the rate** Pump powers in econ() are constants that do not move with rate (P-100 90, P-101 160, P-102 30 kW). With the model's 3 bar pump ΔP, P-101 needs ≈ 65.5 kW and P-102 ≈ 4.5 kW (η assumed).	model econ() · LL-001 · Process engineering (pump hydraulics)	Process engineering to issue pump hydraulics (rated flow, head, NPSH); the load list keeps the larger model values until then.	**OPEN**	ep-10 ee-1 ee-2 ee-5
EQ-ee-4	Safety	Electrical / Safety	**R-101 boiling ceiling (258 °C) is above the cyclohexane auto-ignition temperature (≈ 245 °C)** The boiling ceiling of R-101 (TbC at PI-101 ≈ 258.3 °C) is above the auto-ignition temperature of cyclohexane (≈ 245 °C, typical literature value). Only TSHH-100 at 220 °C keeps the reactor and pump-around surfaces below it.	HAC-001 · Safety layers (critical limits) · SRS (SIF-1)	Keep R-101 and pump-around insulation and cladding intact (no oil-soaked lagging); record hot-surface ignition as a consequence in the HAZOP / LOPA of the runaway scenario; electrical equipment stays T3.	**OPEN**	em-1 er-1 er-6
EQ-ee-5	Design	Electrical / Mechanical	**Agitator power can exceed the 200 kW motor step; heat balance and JI-101 use different powers** The model lets the agitator power reach 204.3 kW (ungassed, $n_1 \geq 1.3\ n_1 d$) above the 200 kW IEC step; the heat balance in step() uses 180 kW while JI-101 reads 153 kW gassed.	model step() / measure() · LL-001 · Mechanical (M-101 datasheet)	M-101 rated 250 kW until the agitator datasheet gives the maximum ungassed power at the highest level and the coldest start-up.	**OPEN**	ep-10 ee-1 ee-2 ee-3
EQ-ee-6	Safety	Electrical / Instrumentation	**EDV-101 fail-open with a de-energise-to-trip SIS depressures the loop on loss of SIS power** EDV-101 is assumed fail-open and the SIS de-energises to trip: a loss of SIS power would open EDV-101 and depressure the hydrogen loop to the flare, although only the ESD opens it in the model.	Safety layers (SL_ASSUMED) · SRS · C&E · LL-001 UPS list	Decide in the SRS whether loss of SIS power is an ESD (EDV-101 opens) or a unit trip (EDV-101 energise-to-open with line monitoring, or held shut on a separate supply).	**OPEN**	er-7 ei-4
ID	**Priority**	**Raised by / owner**	**Query**	**Documents involved**	**Proposed resolution**	**Status**	**Same topic**
EQ-ee-7	Design	Electrical / Electrical	**P-101 ride-through (≈ 2 min 50 s) relies on A/B pumps on separate bus sections**	Reliability tab · LL-001 · single-line diagram (not	Show P-101A and P-101B (and the other A/B pairs) on different LV bus sections	**OPEN**	

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
			Loss of the running P-101 leaves about 2 min 50 s before TSHH-100 (Reliability tab run). An auto-start of P-101B only helps if A and B are on separate bus sections and the bus survives the event.	issued)	in the single-line diagram; add re-acceleration for dips.		
EQ-ec-1	Design	Procurement to commissioning / Process	**How R-102 keeps the Raney-Ni suspended is not defined** Both slurry reactors need the Raney-Ni kept in suspension, but the model has an agitator motor only on R-101 (MOTORS: M-101); the P&IDs draw M-102 on R-102 (EQ-px-4) while the model and the load list have none. How R-102 keeps the catalyst suspended is not defined.	PK-01; equipment list CHX-V1-ME-EQL-001; load list CHX-V1-EE-LL-001	Process to state the R-102 mixing (agitator or gas-induced). If agitated, add M-102 to PK-01, the load list and the motor solo runs.	**OPEN**	em-7 px-4
EQ-ec-2	Safety	Procurement to commissioning / Process	**MP steam on the E-101 shell vs PSV-109: no agreed shell / drum design pressure for test pack TP-05** In HEAT mode (HS-101) MP steam at 10 bar g reaches the E-101 shell, which is the D-101 steam side protected by PSV-109 at an assumed 8.0 bar g (Safety layers SL-R12). The shell, the drum and TP-05 have no agreed design pressure.	PK-03, PK-04, TP-05; Safety layers SL-R12; relief load basis CHX-V1-PE-RLB-001	Either design the E-101 shell and D-101 for the MP steam pressure (PSV-109 set to suit) or add pressure reduction and relief on the MP steam to E-101. Then fix TP-05.	**OPEN**	ep-3 em-2 er-4 px-2 ej-9
EQ-ec-3	Documentation	Procurement to commissioning / Mechanical	**The Raney-Ni catalyst pot has no equipment number or datasheet** The Raney-Ni catalyst pot on R-101 (HV-117) has no equipment number on the P&ID and no datasheet, yet SOP-CYC-07 pressures it with nitrogen above the R-101 pressure.	P&ID CHX-V1-PID-001; PK-05; SOP-CYC-07	Give the pot an equipment number and a datasheet; design pressure at least the loop design pressure; add it to test pack TP-01.	**OPEN**	em-8 ec-5 ej-6
EQ-ec-4	Safety	Procurement to commissioning / Instrumentation	**The SIS scope is not frozen** The SIS scope is not frozen. Open recommendations add sensors, logic and valves: HAZOP R01, R03–R07, R14, R16, R19, R24 and R26; Safety layers SL-R3 … SL-R7; the second make-up valve of the LOPA and SRS.	PK-10; C&E CHX-V1-IN-CE-001; SRS CHX-V1-IN-SRS-001	Close these in the HAZOP close-out before the SIS PO (week 18 in part 4). Order with spare I/O (about 20 %, typical). Re-issue the C&E, then the SIS FAT procedure.	**OPEN**	ei-7 ej-7
EQ-ec-5	Safety	Procurement to commissioning / Process	**Water-wet Raney-Ni: no water-displacement step before charging** Raney-Ni usually comes water-wet. SOP-CYC-07 rinses with cyclohexane but has no step to replace the water. Water charged into R-101 at 200 °C flashes, then ends up in V-104 and the product.	PK-13; SOP-CYC-07; checklist CL-03	Specify the carrier liquid in PK-13: either supplied under cyclohexane, or a solvent exchange in the pot as a written step in SOP-CYC-07.	**OPEN**	em-8 ec-3 ej-6
EQ-ec-6	Documentation	Procurement to commissioning / Process	**PSV sets are assumed and orifices unknown when the relief valves are ordered** The relief valves (SL-R8 … SL-R16) are recommended additions with assumed set pressures and no orifice sizes. PK-15 cannot be bought, and MC-04 and PSSR item PS-03 cannot be closed, until the relief load basis is issued.	PK-15; MC-04; PS-03; Relief & vent tab CHX-V1-PE-RLB-001	Issue the relief load basis before week 24 (the PK-15 PO week); set pressures follow the design pressures in EQ-ec-2.	**OPEN**	la-3 la-8

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-ec-7	Design	Procurement to commissioning / Mechanical	**Benzene feed line design pressure assumed equal to the loop for test pack TP-04** The design pressure of the benzene feed line (TP-04) is assumed equal to the loop (36 bar g). The P-100A/B shut-off head with a closed outlet may be higher.	PK-06; TP-04; line list CHX-V1-PE-LL-001	Take the P-100 shut-off pressure from the pump datasheet. Set the line design pressure, and the TP-04 test pressure, from it.	**OPEN**	ej-5
EQ-px-1	Documentation	EPC P&ID / Project	**Screen drawing and EPC sheets carry the same number CHX-V1-PID-001 rev 0** Both the screen drawing title block and these sheets carry "CHX-V1-PID-001 rev 0".	CHX-V1-PID-001; screen P&ID title block (ui1.js)	keep the number, the sheets' rev table names the screen drawing as current; or give the screen drawing "sheet 0" in ui1.js finishPid (lead's file).	**OPEN**	la-7 ej-1 ej-2 ej-10
EQ-px-2	Safety	EPC P&ID / Instrumentation	**The model trip closes a hand valve (HV-110); the EPC P&ID adds HY-101D** The model closes the hand valve HV-110 on a unit trip (bmsTrip); an SIS cannot close a hand valve. The sheet adds HY-101D (FC) on the MP steam line. Model unchanged.	CHX-V1-PID-001 sheet 1; model bmsTrip(); CHX-V1-IN-CE-001	Keep HY-101D (FC) as the trip-closed valve on the MP steam line; the model's trip closes the hand valve HV-110 — model owner to close HY-101D instead (lead)	**OPEN**	ep-3 em-2 er-4 ec-2 ej-9
EQ-px-3	Documentation	EPC P&ID / Process	**Make-up hydrogen tie-in: PFD into V-104, P&IDs into the K-101 suction** PFD CHX-V1-PFD-001 draws make-up H2 into V-104; screen drawing and these sheets put it into the K-101 suction.	CHX-V1-PFD-001; CHX-V1-PID-001 sheet 2; CHX-V1-PE-LL-001 6"-H-2010-B1	Process decides the tie-in (K-101 suction, as drawn on the P&ID and the screen drawing); PFD rev 1 and LL rev 1 follow	**OPEN**	ep-1 em-6 er-8 px-5 la-5 ej-8
EQ-px-4	Design	EPC P&ID / Process	**M-102 is drawn on both P&IDs but is not a motor in the model** M-102 (R-102 agitator) is drawn on the screen drawing and here, but is not a motor in the model (MOTORS).	CHX-V1-PID-001 sheet 2; model MOTORS; CHX-V1-ME-EQL-001; CHX-V1-EE-LL-001	Process confirms how R-102 keeps the catalyst suspended (EQ-em-7); then the model, the load list and the C&E follow	**OPEN**	em-7 ec-1
EQ-px-5	Documentation	EPC P&ID / Process	**FV-104 is drawn two-way on the screen P&ID, three-way in its note and in the model** Screen drawing shows FV-104 two-way while its note 4 (and the model) say three-way; drawn three-way here.	screen P&ID (ui1.js); CHX-V1-PID-001 sheet 1	Screen drawing to show FV-104 as three-way, as its note 4 and the model (lead, ui1.js)	**OPEN**	ep-1 em-6 er-8 px-3 la-5 ej-8
EQ-la-1	Design	Layout & isometrics / Process	**P-101 NPSH: the line list assumed 4.0 m of static head** Process engineering computed NPSHa ≈ 3.9 m with R-101 bottom only 4.0 m above the P-101 centre line (EP_AS.hNPSH, assumed). The plot plan puts the R-101 bottom at EL +7.0 and the P-101 centre line at EL +0.9: 6.1 m of static head, NPSHa as part 2 shows (bubble-point liquid, no vapour-pressure credit).	CHX-V1-PE-LL-001 (part 6 hydraulics), CHX-V1-ME-DS-P-101, this plot plan	LL rev 1 adopts the layout elevations; the P-101 datasheet states NPSHr ≤ 4.0 m at the rated 998 m³/h (assumed here) or the reactor is raised further.	**OPEN**	ei-9
EQ-la-2	Documentation	Layout & isometrics / Process	**Relief basis uses assumed elevations** The relief load basis took bottom tangents R-101 4.0, R-102 2.0, R-103 1.5, V-104 1.0, V-103 9.0, D-101 8.0 m (assumed) for the fire case (wetted area up to 7.6 m above the fire level). The plot plan gives R-101 7.95, R-102 5.8, R-103 2.49, V-104 CL 2.6, V-103 15.3, D-101 CL 9.6 m; R-101 and R-102 stand on a structure with a solid floor at EL +8.0 only if the floor is plated (here: grating, so the fire level stays at grade).	CHX-V1-PE-RLB-001 (AS.geom), this plot plan, EQ-er-10	Relief & vent re-runs rlb_data.js with layout.json elevations (closes EQ-er-10); the R-101 and R-102 fire loads fall because less shell lies within 7.6 m of grade.	**CLOSED** Closed by CHX-V1-PE-RLB-001 rev 1 together with EQ-er-10 (layout.json elevations and lengths; header 176 m, PSV-101 tailpipe 20.5 m, PSV-109 vent 9.9 m). The plot plan's copy of the RLB refreshes when the Layout tab re-runs.	er-10 la-9

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-la-3	Safety	Layout & isometrics / Process	**PSV tailpipe and header lengths differ from the relief basis** The RLB assumed 20 m tailpipes, 1.5 m inlets and a 150 m unit header to V-190. On the plot plan the tailpipes, inlets and header have the lengths of part 3 (ISO-05). PSV-104 and PSV-112 are raised above the header so their tailpipes drain to it, which lengthens their inlet lines; the header is longer than 150 m because V-190 and the flare sit outside the tank-farm spacing.	CHX-V1-PE-RLB-001 (AS.tail, AS.inlet, AS.hdr), this plot plan	Relief & vent re-checks inlet loss (3 % rule) and built-up back pressure with the lengths of part 3; a 30" header number is proposed for LL rev 1 (EQ-er-9).	**CLOSED** Closed by CHX-V1-PE-RLB-001 rev 1 together with EQ-er-10 (layout.json elevations and lengths; header 176 m, PSV-101 tailpipe 20.5 m, PSV-109 vent 9.9 m). The plot plan's copy of the RLB refreshes when the Layout tab re-runs.	ec-6 la-8
EQ-la-4	Documentation	Layout & isometrics / Piping	**Flare header and PSV tailpipes have no line-list numbers** The isometric of the relief system uses P&ID numbers 6"-FL-1040 and 8"-FL-2016 (both HOLD sizes: the RLB gives 24", 10", 8", 8") and the line-list numbers of the EDV-101 / HV-121 lines; the 30" unit header has no number anywhere.	CHX-V1-PE-LL-001, CHX-V1-PID-001 sheets 1–2, CHX-V1-PE-RLB-001	LL rev 1 adds the tailpipes and 30"-FL-3015-A2 (proposed here); P&ID rev 1 releases the HOLD sizes.	**OPEN**	em-4 em-5 er-9 ei-6 ej-3
EQ-la-5	Documentation	Layout & isometrics / Process	**Make-up hydrogen tie-in point** The plot plan follows the EPC P&ID (make-up into the K-101 suction, station beside the compressor shelter). The line list (6"-H-2010-B1 → V-104) and the PFD tie it into V-104 (EQ-px-3).	CHX-V1-PID-001-2, CHX-V1-PE-LL-001, CHX-V1-PFD-001	Process decides; LL rev 1 or P&ID rev 1 follows; the layout station does not move.	**OPEN**	ep-1 em-6 er-8 px-3 px-5 ej-8
EQ-la-6	Design	Layout & isometrics / Mechanical	**K-101 suction line cannot both drain to V-104 and enter a top nozzle** The line list asks 10"/8"-H-2007 to slope back to V-104 (no liquid to K-101). With K-101 on a block foundation the suction nozzle (top, EL +3.8) is below the line run (EL +5.6), so the last 1.9 m drops into the machine.	CHX-V1-PE-LL-001, CHX-V1-ME-DS-K-101, this plot plan	K-101 vendor to confirm nozzle orientation; a suction strainer with a low-point drain upstream of the drop, or a table-top mounting with down-facing nozzles.	**OPEN**	ep-8
EQ-la-7	Documentation	Layout & isometrics / Mechanical	**Tank sizes are not issued** TK-100, TK-106 and TK-107 are outside the battery limit and have no datasheet; the plot plan sizes them for 48 h of storage (assumed) to check spacing and the bund.	this plot plan, CHX-V1-PM-DOC-001 (not issued)	Offsites issues tank datasheets; the bund height is re-checked (110 % of the largest tank).	**OPEN**	px-1 ej-1 ej-2 ej-10
EQ-la-8	Safety	Layout & isometrics / Process	**PSV-109 vent height set by the platforms nearby** PSV-109 discharges steam to atmosphere. The RLB assumed a 4 m tailpipe; the layout needs the vent outlet 3 m above the highest platform within 15 m (typical practice), so the vent stack is taller.	CHX-V1-PE-RLB-001 (AS.tail109), this plot plan	Relief & vent re-checks PSV-109 built-up back pressure with the vent length of part 2.	**CLOSED** Closed by CHX-V1-PE-RLB-001 rev 1 together with EQ-er-10 (layout.json elevations and lengths; header 176 m, PSV-101 tailpipe 20.5 m, PSV-109 vent 9.9 m). The plot plan's copy of the RLB refreshes when the Layout tab re-runs.	ec-6 la-3

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
EQ-la-9	Safety	Layout & isometrics / Safety	**Control-room siting relies on distance only** CR-101 is not blast-rated; it sits more than 60 m from the process and upwind of the SW monsoon (typical practice). During the NE monsoon it is downwind of the unit.	this plot plan, HAZID (H19 congestion review, H28), CHX-V1-EE-HAC-001	A facility-siting study (explosion overpressure at CR-101), positive-pressure HVAC and gas detection at the air intake.	**OPEN**	er-10 la-2
EQ-ej-1	Documentation	Project & deliverables / Instrumentation	**The SRS CHX-V1-IN-SRS-001 has a number but no issued document** The SIS package PK-10, the C&E CHX-V1-IN-CE-001 and the commissioning plan (proof tests, SIS function tests) cite the SRS by number; it exists only as the SRS tab and as the "srs" data inside sf.json. A vendor cannot buy or FAT a logic solver against a web page.	CHX-V1-IN-SRS-001; CHX-V1-PC-PKG-001 PK-10; CHX-V1-IN-CE-001; CHX-V1-CM-PLN-001	Issue the SRS as a PDF from sf.json (same generator pattern as the C&E) before the SIS purchase order (week 18 in the package schedule).	**CLOSED** Closed: CHX-V1-IN-SRS-001 issued as a PDF (Rev 0, 29 pages, A4 landscape) from the same data as the SRS tab (sf.json, the model process-safety-time runs, the LOPA block diagrams).	px-1 la-7 ej-2 ej-10
EQ-ej-2	Documentation	Project & deliverables / Project	**Deliverables the documents rely on have not been issued** The documents assume a plot plan (relief elevations, fire-water positions, cable lengths, hazardous-area extents), a piping material specification (classes A1 … B3), a single-line diagram (bus sections for A/B motors), instrument and analyser datasheets, a catalyst specification and a fire & gas C&E. None is in this register; the check "every deliverable the documents rely on is issued" lists who cites each.	CHX-V1-PE-RLB-001; CHX-V1-PE-LL-001; CHX-V1-EE-LL-001; CHX-V1-EE-HAC-001; CHX-V1-PC-PKG-001	Add each to the deliverable list with an owner and a planned issue date; until each is issued the documents keep their assumptions labelled, and each issue is followed by a re-run of the documents that cite it (as the relief load basis did for the plot plan).	**PARTLY CLOSED** Plot plan issued as CHX-V1-PL-PP-001 (Rev 0, 28 pages, A3 landscape); the relief load basis re-ran on it (EQ-er-10). Still not issued: Piping material specification (classes A1 … B3); Single-line diagram; Instrument and analyser datasheets; Catalyst specification; Fire & gas cause-and-effect.	px-1 la-7 ej-1 ej-10
EQ-ej-3	Documentation	Project & deliverables / Piping	**The EPC P&ID and the line list use the same sequence numbers for different lines** 26 sequence numbers appear in both documents and 18 of them name a different line: e.g. 6"-P-1003 on the P&ID is the start-up cyclohexane (B/L) line, while 18"-P-1003-B3 in the line list is R-101 bottom → P-101A/B suction. The instrument list locates instruments by line-list numbers, the datasheet nozzle schedules carry both; a field query quoting "line 1003" is ambiguous.	CHX-V1-PID-001; CHX-V1-PE-LL-001; CHX-V1-IN-INL-001; CHX-V1-ME-DS-001	One numbering scheme for the unit: the P&ID rev 1 takes the line-list numbers (size-service-sequence-class); the old P&ID numbers are listed once as superseded (with EQ-em-4, EQ-ei-6).	**OPEN**	em-4 em-5 er-9 ei-6 la-4
EQ-ej-4	Design	Project & deliverables / Process	**E-106 cooling-water flow has three different values** Line list 8"-CW-3009-A1 and the E-106 datasheet use 60 kg/s (econ()); the EPC P&ID line 10"-CW-2030 is sized for 83.5 kg/s (duty/(cp·10 K)); EQ-ep-7 proposes at least 77 kg/s to keep the return ≤ 45 °C. Pipe size, exchanger rating and the CW header load differ with the value chosen.	CHX-V1-PE-LL-001; CHX-V1-ME-DS-001 (DS-E-106); CHX-V1-PID-001 sheet 2; CHX-V1-PE-HMB-001	Process fixes the E-106 CW flow (with EQ-ep-7); LL, datasheet and P&ID are revised to it.	**OPEN**	ep-7
ID	**Priority**	**Raised by / owner**	**Query**	**Documents involved**	**Proposed resolution**	**Status**	**Same topic**
EQ-ej-5	Safety	Project & deliverables /	**Test-pack pressures use the loop design pressure where the EQL and the line list design higher**	CHX-V1-CM-PLN-001 TP-01 … TP-06;	Split the packs at design-pressure breaks or test the higher-design items	**OPEN**	ec-7

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
		Mechanical	The commissioning plan sets every pack from SL_ASSUMED (36 / 8 / 10 bar g), but the EQL and the line list design F-102A/B 40 bar g (in TP-01), P-101 and the E-101 tubes 40 bar g (TP-03), P-100 and 3"-P-1002-B1 40 bar g (TP-04, the open question of EQ-ec-7 is answered by the line list), the E-101 shell 12 bar g (TP-05) and the R-102 jacket 12 bar g (TP-06). Those items would be tested below 1.3 × (or 1.5 ×) their own design pressure.	CHX-V1-ME-EQL-001; CHX-V1-PE-LL-001	separately (shop test); recompute TP-01, -03, -04, -05, -06 from the EQL and line list.		
EQ-ej-6	Documentation	Project & deliverables / Project	**D-102 is assigned to package PK-01 by the EQL but is not in PK-01's scope** The equipment list gives D-102 (Raney-Ni catalyst pot, ID 1,000 × T/T 1,500 mm (assumed)) a datasheet CHX-V1-ME-DS-D-102 and package PK-01; PK-01 lists only R-101, M-101 and R-102. EQ-ec-3 ("the catalyst pot has no equipment number or datasheet") is answered by the EQL but the package was not revised.	CHX-V1-ME-EQL-001; CHX-V1-PC-PKG-001 PK-01; EQ-ec-3	PK-01 (or PK-05) rev 1 adds D-102 with its datasheet; close EQ-ec-3 with a reference to DS-D-102.	**OPEN**	em-8 ec-3 ec-5
EQ-ej-7	Safety	Project & deliverables / Safety	**LOPA recommendation LP-R1 (second make-up valve XV-105B) is in no other document** LP-R1 closes the LOPA gap of SIF-3 PSHH-101 (L3) and of the hydrogen permissive (L4). Only the instrument list carries it (as a recommended row). It has no PSSR disposition in the commissioning plan, is not in the PK-10 tag list, is named only in the notes of the EPC P&ID (not drawn) and has no C&E column.	LOPA tab; CHX-V1-IN-SRS-001; CHX-V1-CM-PLN-001; CHX-V1-PC-PKG-001 PK-10; CHX-V1-PID-001 sheet 2; CHX-V1-IN-CE-001	Approve or reject LP-R1; if approved: P&ID rev 1, C&E column XV-105B, INL rows, PK-10 scope and a "before start-up" PSSR disposition.	**OPEN**	ei-7 ec-4
EQ-ej-8	Documentation	Project & deliverables / Process	**PFD rev 1 is needed to close several queries at once** Four queries ask for the same drawing to change: missing streams 10B and BD (EQ-ep-1), equipment sizes (EQ-em-6, EQ-er-8; the check also finds R-102 drawn Ø3.0 m against 3.2 m in the EQL) and the make-up tie-in (EQ-px-3, also followed by the line list).	CHX-V1-PFD-001; CHX-V1-ME-EQL-001; CHX-V1-PE-LL-001; CHX-V1-PE-HMB-001	Issue PFD rev 1 from the EQL sizes and the HMB streams after Process decides the make-up tie-in; then HMB and LL rev 1.	**OPEN**	ep-1 em-6 er-8 px-3 px-5 la-5
EQ-ej-9	Safety	Project & deliverables / Process	**One hazard, four queries, three different resolutions: MP steam on the E-101 shell** EQ-ep-3, EQ-em-2, EQ-er-4 and EQ-ec-2 each found that MP steam at 10 bar g reaches the E-101 shell, whose water side is protected by PSV-109 at 8 bar g in COOL. They propose different fixes: D-101 and the shell to 12 bar g with PSV-109 re-set; a pressure-reducing valve with its own PSV on the MP steam line; or the existing HAZOP R16 interlock. No discipline owns the decision, so each document carries its own assumption (EQL shell 12, D-101 8, TP-05 at 8).	CHX-V1-PE-LL-001; CHX-V1-ME-EQL-001; CHX-V1-PE-RLB-001; CHX-V1-CM-PLN-001; HAZOP R13/R16; SL-R12	Process owns one decision (worked example in part 6); the other three queries are closed against it.	**OPEN**	ep-3 em-2 er-4 ec-2 px-2
ID	**Priority**	**Raised by / owner**	**Query**	**Documents involved**	**Proposed resolution**	**Status**	**Same topic**
EQ-ej-10	Safety	Project & deliverables /	**HAZOP was made on the screen P&ID; the EPC P&ID adds safeguards it never studied**	CHX-V1-SA-HZP-001; CHX-V1-PID-001; Safety	HAZOP revalidation (or MOC review) of the EPC P&ID before it is issued for	**OPEN**	px-1 la-7 ej-1 ej-2

ID	Priority	Raised by / owner	Query	Documents involved	Proposed resolution	Status	Same topic
		Safety	The HAZOP report (R02) states that relief valves are not shown on the P&ID it studied. The EPC P&ID rev 0 draws PSV-101/102/104/109/112, four TSVs, four EIVs, check valves and the HY change-over valves as recommended items (*). New devices can create new deviations (blocked PSV inlet, EIV closure, change-over valve mis-sequence).	layers tab	construction; record the session in the HAZOP report rev 1.		

3 · Open recommendations of the safety studies

HAZOP R01–R38, Safety layers SL-R1–SL-R44 and LOPA LP-R1. Timing and category are the PSSR disposition of the commissioning plan CHX-V1-CM-PLN-001 (SIS / HW hardware / DOC study or procedure / POST after start-up with an interim measure). "Documents to change" is engineering judgement; "carried by" is computed: the recommendation's new item is drawn on the EPC P&ID, is an instrument-list row, a relief device of the RLB, a package tag or a C&E row. All are OPEN.

Source	before start-up	after start-up, interim measure	no disposition	Open
HAZOP	33	5		38
Safety layers	44			44
LOPA			1	1

Documents to change, by number of recommendations: P&ID 42 · INL 35 · Packages 35 · C&E 29 · SRS 17 · RLB 16 · Plot plan 15 · Line list 12 · SOPs 10 · EQL 9 · F&G C&E (not issued) 7 · HAC 6 · Datasheets 3 · Load list 2 · CM plan 2 · PFD 1

ID	Source	Recommendation	By / layer	PSSR timing	Cat.	Package	Documents to change	Carried by
R01	HAZOP	Add a high-high benzene flow or benzene-to-hydrogen ratio trip (FI-101 against FI-103) that closes XV-101; the feed can run far above design with no trip.	Instrument / Process	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID, Packages	no tag yet
R02	HAZOP	Confirm and document the pressure relief of R-101, R-102, R-103 and V-104 for blocked outlet, make-up valve wide open and fire; relief valves are not shown on the training P&ID.	Process	before start-up	DOC	PK-15	RLB, P&ID, Line list	P&ID, RLB, PKG
R03	HAZOP	Add a high-high temperature trip on R-102 with its own element (set below the R-102 boiling point at loop pressure).	Instrument / Safety	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	no tag yet
R04	HAZOP	Add low-low agitator power (JI-101) to the SIS so that a running motor with a failed drive is seen like a stopped motor.	Instrument / Safety	before start-up	SIS	PK-10	C&E, SRS, INL	P&ID, INL
R05	HAZOP	Add an independent R-101 level element with high-high and low-low actions (stop feed; protect P-101A/B).	Instrument	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	INL
R06	HAZOP	Add low cooling-water flow on E-103 or high-high TI-103 to stop the benzene feed.	Process / Instrument	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	no tag yet
R07	HAZOP	Add independent high-high and low-low level on R-102 (stop P-102A/B, stop feed).	Instrument	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	no tag yet
R08	HAZOP	Fit a non-return valve on the P-100A/B discharge and close XV-101 when no feed pump runs.	Process / Mechanical	before start-up	HW	PK-06	P&ID, Line list, C&E	P&ID
R09	HAZOP	Analyse every benzene receipt for sulphur and water; drain TK-100 water; consider a guard bed ahead of R-101.	Process / Operations	before start-up	DOC	—	SOPs, PFD	no tag yet
R10	HAZOP	Auto-start the standby pumps P-100B, P-101B and P-102B on low discharge pressure or flow.	Electrical / Instrument	after start-up, interim measure	POST	—	INL, Load list, P&ID	no tag yet
R11	HAZOP	Add a low-flow alarm on the pump-around FI-104 (none in the alarm list).	Instrument	before start-up	SIS	PK-11	INL, C&E	no tag yet
R12	HAZOP	Add a control-margin alarm on the FIC-104 output and trend the E-101 heat-transfer coefficient.	Process / Instrument	after start-up, interim measure	POST	—	INL	no tag yet
R13	HAZOP	Confirm the D-101 steam-side relief for blocked export at maximum E-101 duty.	Process	before start-up	DOC	PK-15	RLB	P&ID, RLB, PKG
R14	HAZOP	Add low-low and high-high level on D-101 independent of LIC-109.	Instrument	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	INL

ID	Source	Recommendation	By / layer	PSSR timing	Cat.	Package	Documents to change	Carried by
R15	HAZOP	Monitor D-101 blowdown or condensate for hydrocarbon (E-101 tube leak).	Process	after start-up, interim measure	POST	—	INL, P&ID	no tag yet
R16	HAZOP	Interlock HV-110 (MP steam to E-101) closed whenever XV-101 is open, and put the HEAT to COOL switch-over in the light-off steps.	Instrument / Operations	before start-up	SIS	PK-10	C&E, INL, P&ID, SOPs	P&ID, INL
R17	HAZOP	Detect hydrogen and hydrocarbon in the CW return of E-103, E-106 and the R-102 jacket (degassing pot or analyser).	Process	before start-up	HW	PK-12	INL, P&ID, HAC	no tag yet
R18	HAZOP	Route product to TK-107 automatically on AI-103 or AI-104 high and alarm a stale analyser value.	Instrument / Operations	after start-up, interim measure	POST	—	INL, P&ID	no tag yet
R19	HAZOP	Add an independent low-low level on V-104 closing a shutdown valve in the product line; check the TK-106 / TK-107 vents for gas blow-by.	Process / Instrument	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID, Packages	P&ID, INL
R20	HAZOP	Add a low ΔP alarm on PDI-102 (torn candle) and a routine fines check of the product.	Instrument / Operations	after start-up, interim measure	POST	—	INL, C&E, SOPs	no tag yet
R21	HAZOP	Add a PI-101 high alarm between the PIC-105 alarm and PSHH-101, because PIC-105 measures at V-104, not at R-101.	Instrument	before start-up	SIS	PK-11	INL, C&E	INL
R22	HAZOP	Add a high alarm on TI-109 (V-104 temperature).	Instrument	before start-up	SIS	PK-11	INL, C&E	no tag yet
R23	HAZOP	Add a low-flow alarm on the purge FI-107 and a purity check in the shift routine.	Instrument / Operations	before start-up	SIS	PK-11	INL, C&E, SOPs	no tag yet
R24	HAZOP	Add low-low hydrogen flow (FI-103) to the SIS to close XV-101; today only a stopped K-101 motor is seen.	Instrument / Safety	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	INL
R25	HAZOP	Install hydrogen gas detection at K-101, the reactors and V-104, and alarm the loop hydrogen balance (make-up against 3 × benzene fed).	Safety / Instrument	before start-up	HW	PK-17	HAC, INL, Packages, F&G C&E (not issued)	PKG
R26	HAZOP	Add a second loop oxygen analyser (2oo2) for the XV-105 permissive and record the O2 readings in the purge procedure.	Instrument / Operations	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID, Packages	INL
R27	HAZOP	Fit a non-return valve on the K-101 discharge and confirm the kickback / anti-surge protection and its high-temperature alarm.	Mechanical	before start-up	HW	PK-07	P&ID, Datasheets, Packages	P&ID, PKG
R28	HAZOP	Protect the nitrogen connection: non-return valve, double block and bleed or a removable spool; HV-120 opened only below the header pressure.	Process / Mechanical	before start-up	HW	PK-14	P&ID, Line list, SOPs	no tag yet
R29	HAZOP	Depressuring study for EDV-101 (rate, flare load, minimum metal temperature); fail-open actuator and proof test.	Process	before start-up	DOC	PK-10	RLB, Line list	no tag yet
R30	HAZOP	Provide a flare knock-out drum and a continuous flare purge or seal to keep air out.	Process	before start-up	HW	PK-14	EQL, P&ID, RLB, Line list	P&ID
R31	HAZOP	Write a Raney-Ni handling procedure: always wet and under nitrogen, metered charges through HV-117, filter-cake handling.	Operations / Safety	before start-up	DOC	—	SOPs, CM plan	no tag yet
R32	HAZOP	Light-off and shutdown steps: limit the feed ramp after XV-101 opens, stop benzene before hydrogen, and add a low-temperature-with-feed interlock (TI-100 with XV-101 open) to stop benzene accumulation.	Operations / Instrument	before start-up	DOC	PK-10	SOPs, C&E, SRS	no tag yet
R33	HAZOP	Add a TIC-101 against TI-100 deviation alarm (two elements in the same reactor).	Instrument	before start-up	SIS	PK-11	INL, C&E	INL
R34	HAZOP	Gas-free test and positive isolation before opening any equipment (filters, reactors); written filter change-over procedure.	Operations	before start-up	DOC	—	SOPs, CM plan	no tag yet

ID	Source	Recommendation	By / layer	PSSR timing	Cat.	Package	Documents to change	Carried by
R35	HAZOP	Fit minimum-flow bypasses on P-100A/B and P-102A/B.	Mechanical	before start-up	HW	PK-06	P&ID, Line list, Datasheets	no tag yet
R36	HAZOP	Review the blanketing and venting of TK-106 / TK-107 for hydrogen flashing from the product.	Process	before start-up	DOC	—	RLB	no tag yet
R37	HAZOP	Add an R-103 bed-rise alarm (TI-107E minus TI-106) as early warning of benzene slip.	Instrument	before start-up	SIS	PK-11	INL, C&E	no tag yet
R38	HAZOP	Positive segregation of MP steam and CW on the R-102 jacket (interlocked change-over) and jacket relief.	Process / Mechanical	before start-up	HW	PK-14	P&ID, C&E, RLB, Line list	P&ID, INL, RLB, PKG
SL-R1	Safety layers	R-101 pressure pre-alarm on PT-101 · setting H 32.5 bar g (assumed)	Layer 2	before start-up	SIS	PK-11	INL, C&E	INL
SL-R2	Safety layers	Deviation alarm TIC-101 element vs TI-100 element · setting |ΔT| > 5 °C (assumed)	Layer 2	before start-up	SIS	PK-11	INL, C&E	INL
SL-R3	Safety layers	D-101 level low-low → unit trip · setting 10 % (assumed)	Layer 3	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	INL
SL-R4	Safety layers	Agitator shaft-speed switch, voted with the M-101 run signal · setting low speed (assumed 50 % of design)	Layer 3	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	P&ID, INL
SL-R5	Safety layers	Separate SIS elements in R-103 (two per cell, 2oo3 per hot zone) · setting 220 °C	Layer 3	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID	INL
SL-R6	Safety layers	V-104 level low-low closes a new SIS valve XV-104 on the product line · setting 5 % (assumed)	Layer 3	before start-up	SIS	PK-10	C&E, SRS, INL, P&ID, Packages	P&ID, INL
SL-R7	Safety layers	Stop P-100A/B on unit trip · setting on every unit trip	Layer 3	before start-up	SIS	PK-10	C&E, SRS, Load list	INL
SL-R8	Safety layers	Safety valve on the R-101 overhead (no isolation between it and R-101) · setting 36 bar g (assumed design pressure)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Line list, EQL, Packages	P&ID, RLB, PKG
SL-R9	Safety layers	Safety valve on the R-102 off-gas line · setting 36 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Line list, EQL, Packages	P&ID, RLB, PKG
SL-R10	Safety layers	Safety valve on V-104 · setting 36 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Line list, EQL, Packages	P&ID, RLB, PKG
SL-R11	Safety layers	Safety valve on K-101 discharge (upstream of FV-103) · setting 42 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Line list, EQL, Packages	P&ID, RLB, PKG
SL-R12	Safety layers	Safety valve on steam drum D-101 · setting 8 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, EQL, Packages	P&ID, RLB, PKG
SL-R13	Safety layers	Thermal relief on the E-103 cooling-water side · setting 10 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Packages	P&ID, RLB, PKG
SL-R14	Safety layers	Thermal relief on the E-106 cooling-water side · setting 10 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Packages	P&ID, RLB, PKG
SL-R15	Safety layers	Thermal relief on the R-102 jacket · setting 10 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Packages	P&ID, RLB, PKG
SL-R16	Safety layers	Thermal relief on each catalyst filter F-102A/B · setting 36 bar g (assumed)	Layer 4	before start-up	HW	PK-15	RLB, P&ID, Packages	P&ID, RLB, PKG
SL-R17	Safety layers	Emergency isolation valve on benzene at battery limit · setting fail closed, fire-safe	Layer 5	before start-up	HW	PK-15	P&ID, INL, Packages	P&ID, INL, PKG
SL-R18	Safety layers	EIV on the R-101 bottom outlet (pump-around suction) · setting fail closed, fire-safe	Layer 5	before start-up	HW	PK-15	P&ID, INL, Packages	P&ID, INL, PKG
SL-R19	Safety layers	EIV on the R-102 bottom outlet · setting fail closed, fire-safe	Layer 5	before start-up	HW	PK-15	P&ID, INL, Packages	P&ID, INL, PKG
SL-R20	Safety layers	EIV on the V-104 product line · setting fail closed, fire-safe	Layer 5	before start-up	HW	PK-15	P&ID, INL, Packages	P&ID, INL, PKG
SL-R21	Safety layers	Check valve at P-100A/B discharge	Layer 5	before start-up	HW	PK-15	P&ID, Packages	P&ID, PKG
SL-R22	Safety layers	Check valve at K-101 discharge	Layer 5	before start-up	HW	PK-07	P&ID, Packages	P&ID, PKG

ID	Source	Recommendation	By / layer	PSSR timing	Cat.	Package	Documents to change	Carried by
SL-R23	Safety layers	Flare knock-out drum	Layer 5	before start-up	HW	PK-14	EQL, P&ID, RLB, Line list	P&ID
SL-R24	Safety layers	Kerbed area round the benzene pumps to the oily-water sewer	Layer 5	before start-up	HW	PK-14	Plot plan	none yet
SL-R25	Safety layers	Kerbed slab under the reactor structure to a closed drain	Layer 5	before start-up	HW	PK-14	Plot plan	none yet
SL-R26	Safety layers	Raney-Ni handled under liquid and nitrogen (catalyst pot) · setting N2 blanket	Layer 5	before start-up	DOC	PK-13	SOPs, EQL	none yet
SL-R27	Safety layers	Spent catalyst from F-102 kept wet in closed drums	Layer 5	before start-up	DOC	PK-13	SOPs	none yet
SL-R28	Safety layers	Fireproofing of the R-103 skirt and V-104 saddles · setting 2 h (assumed)	Layer 5	before start-up	HW	PK-14	EQL, Datasheets, Plot plan	none yet
SL-R29	Safety layers	Fire-water ring main · setting 10 bar g (assumed)	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
SL-R30	Safety layers	Hydrant west of the reactors	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
SL-R31	Safety layers	Hydrant east of V-104	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
SL-R32	Safety layers	Fire-water monitor on the reactor structure	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
SL-R33	Safety layers	Fire-water monitor at V-104 / K-101	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
SL-R34	Safety layers	Deluge on K-101 · setting opens on confirmed fire (assumed)	Layer 6	before start-up	HW	PK-17	Plot plan, Packages, F&G C&E (not issued)	PKG
SL-R35	Safety layers	Water spray on V-104 · setting opens on confirmed fire (assumed)	Layer 6	before start-up	HW	PK-17	Plot plan, Packages, F&G C&E (not issued)	PKG
SL-R36	Safety layers	Hydrogen detector at K-101 · setting 10 % / 25 % LFL (assumed)	Layer 6	before start-up	HW	PK-17	HAC, Packages, F&G C&E (not issued)	PKG
SL-R37	Safety layers	Hydrogen detector at the R-101 top flanges · setting 10 % / 25 % LFL (assumed)	Layer 6	before start-up	HW	PK-17	HAC, Packages, F&G C&E (not issued)	PKG
SL-R38	Safety layers	Hydrocarbon (benzene) detector at P-100A/B · setting 10 % / 25 % LFL (assumed)	Layer 6	before start-up	HW	PK-17	HAC, Packages, F&G C&E (not issued)	PKG
SL-R39	Safety layers	UV/IR flame detector covering K-101 and V-104	Layer 6	before start-up	HW	PK-17	HAC, Packages, F&G C&E (not issued)	PKG
SL-R40	Safety layers	Manual call point	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
SL-R41	Safety layers	Site siren / PA	Layer 6	before start-up	HW	PK-17	Plot plan	PKG
SL-R42	Safety layers	Windsock	Layer 6	before start-up	HW	PK-17	Plot plan	none yet
SL-R43	Safety layers	Muster point	Layer 6	before start-up	HW	PK-17	Plot plan	none yet
SL-R44	Safety layers	Safety shower and eye wash at the benzene pumps	Layer 6	before start-up	HW	PK-17	Plot plan, Packages	PKG
LP-R1	LOPA	Second make-up shut-off valve XV-105B in series with XV-105: PSHH-101 and the hydrogen permissive with 1oo2 final elements (LOPA L3, L4; SRS SIF-3R, PERM-H2R).	LOPA	no disposition	—	—	P&ID, C&E, SRS, INL, Packages	INL

4 · Checks across the document set

Computed by the same function the page runs (EJ-PURE block of the Project & deliverables tab) on the data of every document. Each failure links to the query that records it.

Group	Check	Result	Detail	Query
Register	Every document in the register has its PDF (page count from pdfinfo)	**PASS**	19 of 19 have a file	
Register	Every document number cited in the documents is in the register (datasheets in the book CHX-V1-ME-DS-001)	**PASS**	32 distinct numbers cited in 15 documents on this page	
Register	Every deliverable the documents rely on is issued	**FAIL**	Piping material specification (classes A1 … B3) (cited by 1: PC-PKG-001); Single-line diagram (cited by 1: EE-LL-001); Instrument and analyser datasheets (cited by 1: PC-PKG-001); Catalyst specification (cited by 1: PC-PKG-001); Fire & gas cause-and-effect (cited by 1: PC-PKG-001)	EQ-ej-2
Equipment	Every process item of the equipment list appears in the HMB (drives excluded)	**FAIL**	14 of 15; not in the HMB: D-102 (batch charge: no HMB stream)	EQ-em-8 EQ-ec-3
Equipment	Every equipment item of the EQL is drawn on the EPC P&ID	**PASS**	17 of 17	
Equipment	Every equipment item on the P&ID sheets is in the EQL	**PASS**	21 P&ID equipment tags	
Equipment	Every PFD item is in the EQL	**PASS**	14 PFD items	
Equipment	PFD equipment sizes = EQL sizes (diameter within 5 %)	**FAIL**	R-101 Ø 3.6 vs 3.80 m; D-101 Ø 1.4 vs 2.40 m (L 4 vs 8.4); R-102 Ø 3 vs 3.20 m; V-104 Ø 2 vs 2.30 m	EQ-em-6 EQ-er-8 EQ-ej-8
Relief	Every PSV/TSV of the relief load basis is set ≤ the design pressure in the EQL of what it protects	**PASS**	PSV-101 36 ≤ R-101 36 · PSV-102 36 ≤ R-102 36 · PSV-104 36 ≤ V-104 36 · PSV-112 42 ≤ K-101 42 · PSV-109 8 ≤ D-101 8 · TSV-113 10 ≤ E-103 10 · TSV-106 10 ≤ E-106 10 · TSV-102J 10 ≤ R-102 jacket 12 · TSV-102F 36 ≤ F-102 40 — set below design: TSV-102J, TSV-102F	EQ-em-11
Relief	Relief-basis fire-case elevations = the plot plan (bottom tangents), lengths taken from the plot plan	**PASS**	R-101 RLB 7.95 · plot plan 7.95 m · R-102 RLB 5.80 · plot plan 5.80 m · R-103 RLB 2.49 · plot plan 2.49 m · V-103 RLB 15.30 · plot plan 15.30 m · from CHX-V1-PL-PP-001: header 176 m, PSV-101 tailpipe 20.5 m, PSV-109 vent 9.9 m	
Relief	Relief sets in the RLB = the Safety-layers register (SL_IT)	**PASS**	9 of 9	
Relief	Every relief device of the RLB is on the EPC P&ID	**PASS**	9 of 9	
Relief	MP steam that can reach the E-101 shell (HS-101 HEAT) stays below the set of PSV-109 protecting it in COOL	**FAIL**	MP steam 8"-S-3002-A1 10.0 bar g vs PSV-109 8.0 bar g (D-101 design 8)	EQ-ep-3 EQ-em-2 EQ-er-4 EQ-ec-2 EQ-ej-9
Instruments	Every trip and alarm cause of the C&E is in the instrument list	**PASS**	35 of 35 trip/alarm rows	
Instruments	Every alarm of an existing instrument in the INL has its C&E alarm row	**PASS**	19 instruments with alarms; recommended alarms waiting for C&E rev 1: PAH-101, TDAH-101	
Electrical	Every motor of the equipment list is in the electrical load list	**FAIL**	5 of 6 motors (M-101, M-102, P-100A/B, P-101A/B, P-102A/B, K-101); missing: M-102	EQ-em-7 EQ-ec-1 EQ-px-4
Electrical	Every load-list motor is in the EQL with the same rating (kW)	**PASS**	8 motor rows	

Group	Check	Result	Detail	Query
Procurement & commissioning	Every package's equipment exists in the EQL	**PASS**	8 equipment packages	
Procurement & commissioning	Every EQL item is in the scope of the package the EQL assigns it to	**FAIL**	15 of 17; outside the package tag list: M-102 → PK-01, D-102 → PK-01	EQ-ej-6 EQ-ec-3 EQ-em-7
Procurement & commissioning	Every datasheet a package cites is in the datasheet book	**PASS**	14 datasheet references, 15 sheets	
Procurement & commissioning	Every test pack is designed for the design pressure of each item and line in it (EQL, line list)	**FAIL**	TP-01 at 36 bar g (test 46.8): higher design F-102A/B 40, 3"-P-1012-B1 40 · TP-02 at 42 bar g (test 46.2) ✓ · TP-03 at 36 bar g (test 46.8): higher design P-101A/B 40, E-101 40, 16"-P-1004-B3 40, 16"-P-1005-B3 40, 16"-P-1006-B3 40, 16"-P-1007-B3 40 · TP-04 at 36 bar g (test 54.0): higher design P-100A/B 40, 3"-P-1002-B1 40 · TP-05 at 8 bar g (test 10.4): higher design E-101 shell 12, 8"-S-3002-A1 12, 3"-BFW-3004-A1 15 · TP-06 at 10 bar g (test 13.0): higher design R-102 jacket 12, 6"-CW-3011-A1 12	EQ-ej-5 EQ-ec-7 EQ-ec-2
Lines	The same line number means the same line on the EPC P&ID and in the line list	**FAIL**	26 sequence numbers used by both; 18 name a different line (e.g. 1003: P&ID 6"-P-1003 (start-up cyclohexane (B/L)) vs LL 18"-P-1003-B3 (R-101 bottom); 1004: P&ID 4"-SL-1004 (D-102 catalyst pot) vs LL 16"-P-1004-B3 (P-101A/B discharge); 1005: P&ID 10"-P-1005 (R-101 overhead) vs LL 16"-P-1005-B3 (FV-104)); same line but another size: 4 (4"-P-1002 vs 3"-P-1002-B1; 10"-P-2005 vs 8"-PG-2005-B2; 10"-G-2007 vs 8"-H-2007-B2; 6"-P-2011 vs 1-1/2"-H-2011-B2); 43 of 47 P&ID labels found on the drawings	EQ-ej-3 EQ-em-4 EQ-ei-6
Lines	E-106 cooling-water flow is the same in the line list, the P&ID and the EQL	**FAIL**	LL 8"-CW-3009-A1 60.0 kg/s · P&ID 10"-CW-2030 83.5 kg/s · EQL E-106 60.0 kg/s	EQ-ej-4 EQ-ep-7
Lines	Nitrogen tie-in has one size in the line list and on the P&ID	**FAIL**	LL 16"-N-3012-A1 (27.7 kg/s, model N_2 flow) vs P&ID 2"-N-2014 (K-101 suction)	EQ-ep-6
Lines	Make-up hydrogen enters the loop at the same point on the PFD, the line list and the P&ID	**FAIL**	PFD → V-104 · LL 6"-H-2010-B1 → V-104 · P&ID 8"-G-2009 → K-101 suction	EQ-px-3 EQ-ej-8
Lines	Outlet lines of vessels, reactors, pumps and filters are designed for at least the item's design temperature	**FAIL**	4"-P-1013-B1 80 °C < V-104 190 °C; 8"-H-2007-B2 80 °C < V-104 190 °C; 1-1/2"-H-2011-B2 80 °C < V-104 190 °C	EQ-em-10 EQ-ep-12
Recommendations	The register holds every HAZOP and Safety-layers recommendation (plus LOPA LP-R1)	**PASS**	82 in HZ_REC + SL_ADD, 83 in the register	
Recommendations	Every open recommendation has a PSSR disposition in the commissioning plan	**FAIL**	82 dispositions for 83 recommendations; none for: LP-R1	EQ-ej-7
Recommendations	The HAZOP basis drawing is the current P&ID	**FAIL**	HAZOP R02 says relief valves are not shown on the P&ID it studied; the EPC P&ID shows 9 (PSV-101, PSV-102, PSV-104, PSV-112, PSV-109, TSV-113, TSV-106, TSV-102J, TSV-102F) among 19 items marked as recommended (*)	EQ-ej-10
Register	Every engineering query in the documents' data is in the EQ register	**PASS**	66 queries in the data on this page, 81 in the register	
Register	HMB design case = the simulator design point (benzene feed FI-101)	**PASS**	HMB 34.80 t/h vs model 34.80 t/h	

5 · Basis and notes

Every document is "Issued for teaching" (Rev 0 unless the register shows a later revision), generated from the Cyclohexane Reactor Ver 1 simulator model by the OTS tab named in the register; originator Apinan Soottitantawat, checker and approver blank. No client, contractor or vendor is implied. Priorities, owners, topics and the documents a recommendation changes are this register's engineering judgement. Phases, revision codes and the query workflow are typical practice. The register's own page count is taken from the previous build of this PDF.